Next, the correlation heatmap, **Figure 3,** below displays multiple distinct patterns that describe quarterback performance. Quarterbacks who throw more passes achieve better results because they complete more passes, attempt more throws, gain more passing yards, score more touchdowns, and achieve more first downs. The heatmap displays a second group of efficiency variables, which includes Y/A, AY/A, and NY/A, ANY/A, and passer rating, because those variables measure quarterback performance in a similar way. The passing efficiency measures from the study exhibit substantial overlap because they measure equivalent aspects of quarterback performance.

The heatmap displays variable movement patterns, which show that different variables followed distinct movement paths. Interception percentage had weaker and, in some cases, negative relationships with efficiency-based measures, which fits the idea that more mistakes tend to lower overall passing quality. The Year and Age variables showed weak relationships with other variables, which indicates they were not essential to the primary performance patterns in the dataset. The heatmap results indicate that quarterback statistics maintained strong relationships with each other, which explains why the initial principal components extracted such a high percentage of data variation.

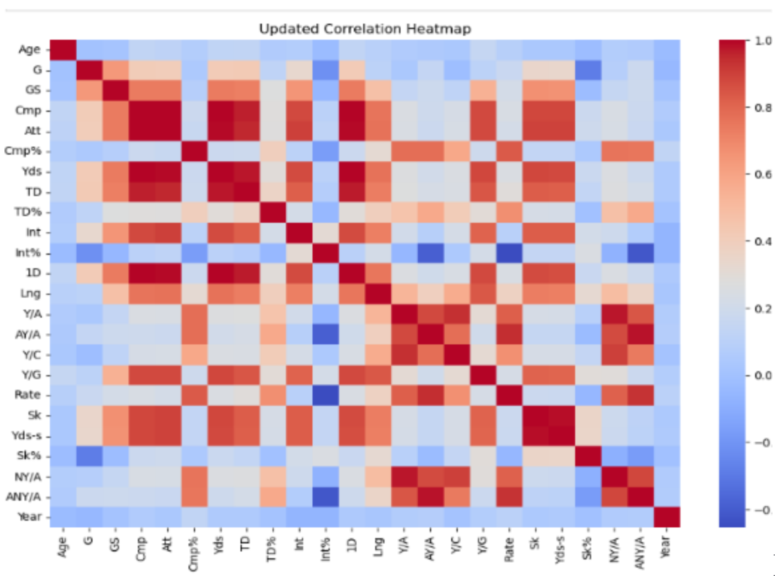


Figure 3

View the figure below: **Figure 4**. It presents the receiver operating characteristic – Area Under Curve (ROC-AUC) analysis, which helps evaluate how well the model distinguishes between quarterbacks who were selected to the Pro Bowl and those who were not. We used this as a secondary measure upon the base analysis, which is covered in the sections above. The model excelled and had an AUC score of 0.959, close to 1, meaning there was good classification performance. If the AUC had fallen below 0.70, that would have pointed to weaker classification performance, while a score near 0.50 would suggest the model was doing little better than random guessing. As shown by the ROC curve, the model kept the false positive rate relatively low while maintaining a strong true positive rate across different thresholds. Overall, this reinforces the results by showing that the model worked well across multiple classification thresholds, not just one.

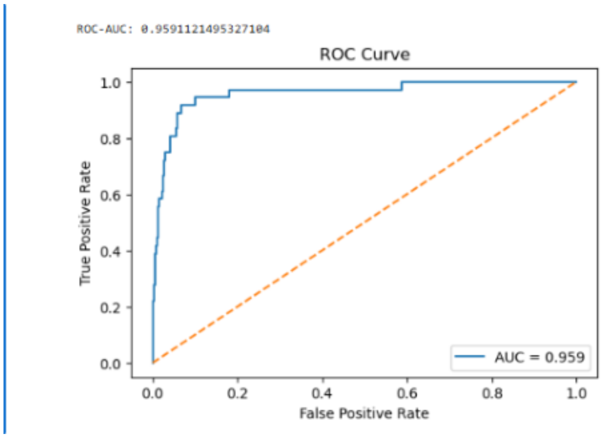
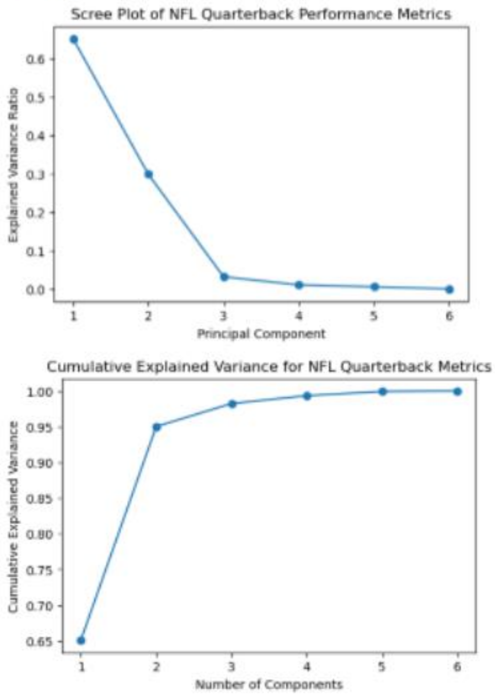


Figure 4

Additionally, the Principal Component Analysis (PCA) helped us examine how much of the variation in quarterback performance could be explained by a smaller number of components [features/variables]. In **Figure 5**, there are two graphs, the first principal component represents overall passing production, while the second reflects passing efficiency. The results demonstrated that the first principal component explained 65.13% of the total variance, and the second principal component explained 29.88% of the total variance. Together, these two components accounted for approximately 95% of the dataset's variation. These insights reveal that most of the information in the quarterback's performance metrics was captured within the first two components, while the remaining components contributed very little additional variation or provided minimal value. The first component of the loading values demonstrated overall passing production because passing yards, completions, touchdowns, and interceptions made strong contributions, but the second component showed passing efficiency because passer rating and ANY/A were its main drivers.

Explained Variance Ratio:
PC1: 0.6513
PC2: 0.2988
PC3: 0.0322
PC4: 0.0110
PC5: 0.0059
PC6: 0.0007



PCA Loadings:

	PC1	PC2	PC3	PC4	PC5	PC6
Yds	0.405722	-0.116952	0.205388	-0.095600	-0.328606	-0.762364
TD	0.488542	-0.083611	0.424681	-0.051297	0.744557	0.130331
Int	0.447327	-0.211378	-0.835990	0.165305	0.170287	0.003906
Cmp	0.493702	-0.126547	0.175878	-0.053673	-0.552969	0.632974
Rate	0.202419	0.670350	0.060764	0.708737	-0.044628	-0.022137
ANY/A	0.175412	0.685031	-0.206839	-0.675053	0.028825	0.025550

Figure 5

Discussion

The study's findings demonstrate that quarterback performance metrics provide essential predictive capabilities that determine Pro Bowl selection outcomes. The logistic regression model achieved 94.61% accuracy, and the ROC-AUC score of 0.959 demonstrated that the model successfully distinguished between selected quarterbacks and non-selected quarterbacks. The results demonstrate that statistical performance serves as a strong basis to assess Pro Bowl-level athletic performance. The selection pattern from the dataset showed that passing production and efficiency metrics were the main factors that explained selection patterns in the dataset.

Furthermore, the model could correctly predict Pro Bowl selection but performed better at identifying players who would not be selected than players who would be selected. The model failed to identify some players who were selected because quarterback statistics did not capture all factors that affected selection.

Pro Bowl recognition depends on multiple factors that extend beyond the on-field performance of players. Fan voting, media attention, team record, popularity, and narrative can all affect who ultimately receives recognition, even when another quarterback may have comparable or stronger statistical results. The study demonstrates that Pro Bowl results cannot be completely understood through performance metrics alone because of external variables within the entertainment sports industry.

However, the implementation of our model can help mitigate errors in the long term. The evaluation maintains consistency through this model because it establishes precise statistical standards that determine Pro Bowl-level quarterback performance. It will serve as a benchmark and help identify the traits most closely tied to true Pro Bowl talent and reduce the chances of outside factors such as reputation or popularity. In that sense, the model can serve as a more objective tool for recognizing quarterbacks whose production and efficiency reflect the level expected of a Pro Bowl player.

To conclude, these findings are useful because they show how predictive modeling can support more objective player evaluation. Teams, analysts, and decision-makers can use performance-based models as an added tool when assessing quarterback value, benchmarking players, or identifying qualified quarterbacks. Still, this study highlights that no model based only on passing data can fully capture the entire Pro Bowl selection process. Future research could strengthen the model by adding variables related to team success, wins, strength of schedule, postseason impact, or public voting influence. Overall, this study shows that quarterback statistics can explain a large share of Pro Bowl selection, but they work best as part of a broader evaluation rather than the only measure.

References

NFL Passing Statistics (2001-2023). (n.d.). Www.kaggle.com.

https://www.kaggle.com/datasets/rishabjadhav/nfl-passing-statistics-2001-2023